

Futures Contracts

Fundamentals of Trading

FINM 37000 - Futures And Related Derivatives

Eric Patterson, Summer 2026

Futures and Forward Contracts

Two Ways to Trade Assets in the Future

Forwards

- Agreement to buy/sell at future time at agreed price
- OTC
- Custom contract specs
- Settled at expiration
 - Gains/losses deferred

Futures

- Agreement to buy/sell at future time at agreed price
- Exchange Traded
- Standardized contract specs
- Settled daily (mark-to-market)
 - Gains/losses assessed daily

Cashflows

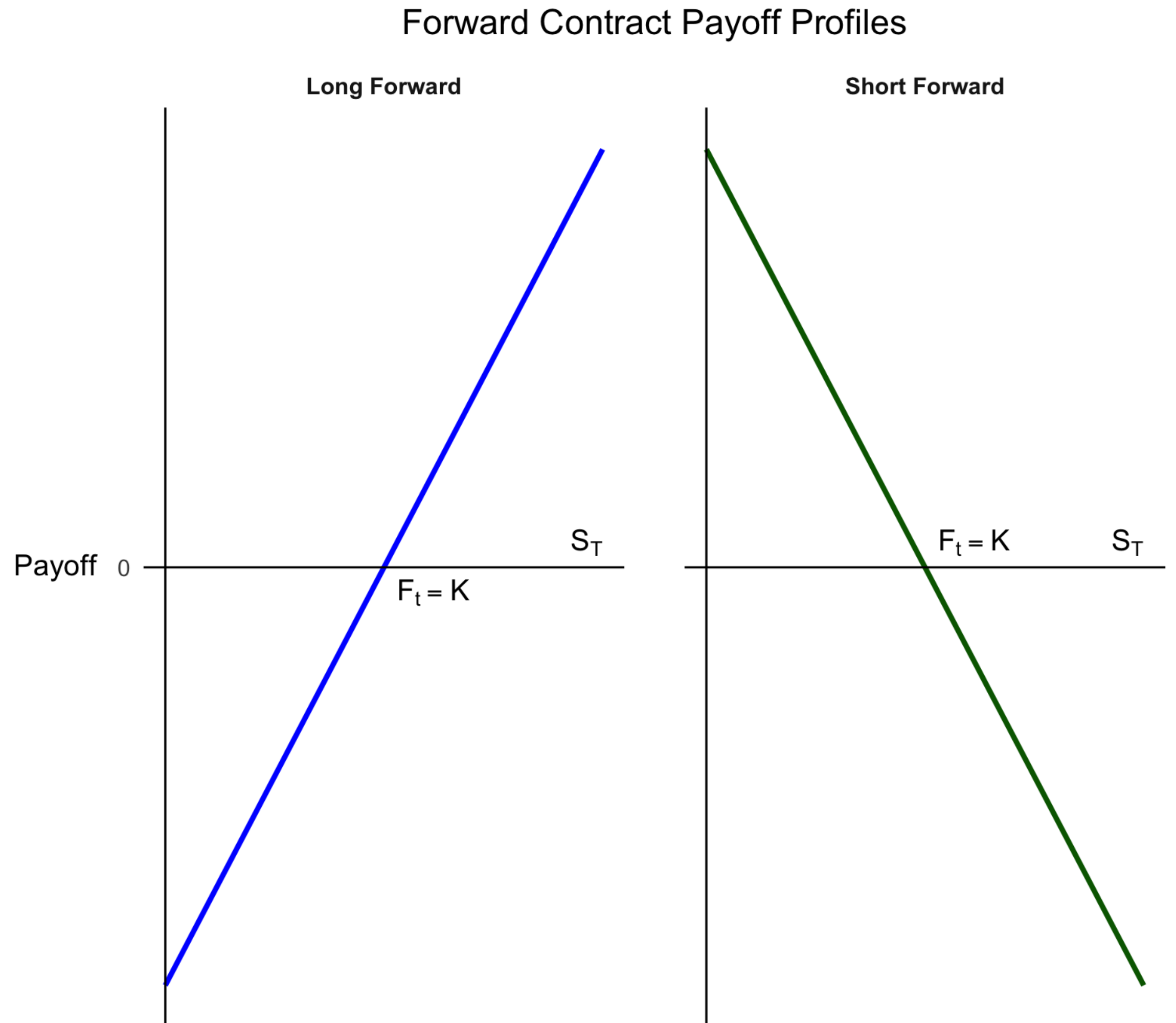
Futures vs. Forwards

- Forward, buy or sell at time T for price K
 - Time of trade: no cashflow
 - Time T : Buyer pays seller K , seller delivers asset
- Future, buy or sell at time T for price K
 - Time of trade: no cashflow between traders, margin accounts must be funded
 - End of trade day: adjust margin accounts (mark to market), position value resets to 0
 - End of next day: mark to market, position value resets
 - ...
 - Time T : mark to market and process delivery if physically settled

Payoff At Expiration

Spot vs. Forward Price

- Spot price (price to buy/sell on the current date t), S_t
- Forward price (price on date t to buy/sell at a future date T), F_t
 - Forward price is NOT the price you pay for a forward contract!!
- Often F_t is written K to emphasize that after entering a forward contract on date t , the contract price does not change.



Mechanics of a futures trade

Example: July 2026 Soybeans (ZSN6)

- 1 Futures contract traded at $1050'2 = 1050.25$
 - Pay \$0 now
 - Fund margin account ([example](#): \$2200/contract)
 - Buyer: If held through expiration, receive soybeans
 - Seller: If held through expiration, deliver soybeans
- Contract Specs
 - Settlement date: 14 July 2026
 - Contract unit: 5,000 bushels
 - Price unit: U.S. cents per bushel
 - Minimum Price Fluctuation: 1/4 cent per bushel

Mechanics of a futures trade

Example: July 2026 Soybeans (ZSN6)

- 1 Futures contract traded at 1050'2
- End of day: settlement price 1060'4 = 1060.50
 - Buyer/long: Margin +10.25¢/per bushel x 5000 bushels = \$512.50
 - Seller/short: Margin -\$512.5

Mechanics of a futures trade

Example: July 2025 Soybeans (ZSN5)

- Futures contract settled at 1060'4
- Next day
 - Buyer exits at a price 1055'4
 - Original Buyer sells
 - No position
 - Margin $-5\text{¢}/\text{per bushel} \times 5000 \text{ bushels} = -\250
 - New buyer: long(er) 1

Open Interest vs. Trade Volume

Definitions

Open Interest: The total number of contracts outstanding.

- Alternatively, the total of all long positions.
- Alternatively, the total of all short positions.
- Requires knowledge of all traders' positions.

Trade Volume: The total number of contracts traded.

- Many trades appear in the data feed from the exchange.
- [Block trades](#) are reported separately.

Open Interest vs. Trade Volume

Example: July 2026 Soybeans (ZSN6)

- Day 1
 - Volume: +1
 - Open Interest: +1, 0, or -1?
- Day 2
 - Volume: +1
 - What happens to open interest?
 - Suppose seller sells another contract to the new buyer
 - Daily volume: +2
 - What happens to open interest?

Final Settlement (Expiration)

Financial Settlement

- No physical delivery.
- Observed price/index/rate determines financial value.
- Examples
 - SOFR
 - S&P 500 Index

Mark to Market

Daily futures cash flows

[Test your understanding](#). Describe cash flow, volume, and open interest for the following events.

- Buy 10 [S&P Minis](#) for \$7680
- Price increases to \$7690.50
- Price increases to \$7700 and you sell 5
- Price settles for the day at \$7675
- Hold your short position until final settlement at expiration for \$7676

Final Settlement (Expiration)

Physical Delivery

- Futures position held through expiration
 - Long: receive delivery of physical
 - Short: deliver physical (warehouse receipt or shipping certificate)

Delivery

Physical Delivery

- Example: July Soybean example: 5000 BU/contract
 - Trades through July 14th (final settlement date)
 - Position Day
 - Short gives exchange intent to deliver between June 29th and July 15th
 - Grades: Yellow #1 (+6¢/BU), Yellow #2, Yellow #3 (-6¢/BU)
 - Locations: 8 different locations +0-16.25¢/BU
 - Exchange selects long to take delivery (time priority)
 - Notification Day (Position+1): Notify long/short they have been matched
 - Delivery Day (Position+2): Long pays short through clearing, receives delivery

The Basis

Futures/Forwards vs. Underlying

- **Basis trade** buys and sells underlying versus futures for potential delivery
- Examples
 - Commodities: grades and locations for delivery
 - Treasuries: Treasury notes with remaining term to maturity meeting specified time range (e.g., 6.5-8 years for Ten year futures), with delivery price adjustment
 - FX: deliver currency
 - Why does it work for financial settlement?

Who Trades, Why, and How Much

- Who
 - hedgers
 - speculators
 - market makers
- Why: Take or reduce risks from
 - Commodities: Grains, Meats, Energy, Metal
 - Fixed income: Treasuries, SOFR, Fed Funds
 - Equity Indexes: S&P, Nasdaq, Dow, Russell, Nikkei
 - FX, Electricity

Who Trades, Why, and How Much

	Soybeans	Corn	Crude	E-mini S&P	Ten Years	3-month SOFR
Contract Size	5000 Bushels	5000 Bushels	1000 Barrels	\$50 x S&P Index	\$100,000 face value at maturity	\$2500 x IMM index
Volume/day	100k-700k	200k-1mill	500k-2.5 mill	1-5 mill	1.5-6 mill	2-15 mill
Open Interest	800-900k	1.5-2 mill	1.5-2 mill	2-3 mill	4-5 mill	10-12 mill

Extreme Events In Futures Markets

Exceptional But Important Concepts

Price limits (going limit up/down): [Maximum allowed price move per day.](#)

- How do you get out of long position when price is limit down? No valid bids.

Short squeeze: Short position demand to make delivery greatly exceeds available supply (not unique to futures markets).

- What happens to physical prices?
- What happens to futures prices?

Extreme Events In Futures Markets

Exceptional But Important Concepts

Cornering the market: Controlling physical supply to induce a short squeeze (not unique to futures markets).

- What position would the corner want to take in futures?
- Many famous examples, most ended poorly.
 - Onion futures corner in the 1950s
 - Lead to Onion Futures Act and CME restructuring
 - Hunt brothers/Silver Thursday, March 27, 1980
 - Exchange changed rules on commodity purchases on margin
 - Sumitomo/Hanamaka Copper market corner in the 1990s
 - \$2.6 billion loss, 8 years in prison